

A Formal Path Dilation for Every Finite-Horizon Evolution Family

Run fa_banach_001, agent_lane_19

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Status. Partial result / clarification, likely valid. The source asks which C_0 -semigroups and C_0 -evolution families admit an approximate invertible dilation. Read literally, without requiring the dilation space to preserve Hilbertianity, 2-smoothness, UMD, or related geometry, the answer is: all finite-horizon C_0 -evolution families admit an exact invertible dilation. The construction is too large for the stochastic maximal inequality theorem in the source, so this is not a solution of the intended geometry-preserving problem.

Source Question

The source is Jan van Neerven and Mark Veraar, *Maximal estimates for stochastic convolutions in 2-smooth Banach spaces and applications to stochastic evolution equations*, arXiv:2006.08325. In Section 6, the authors ask:

“Which C_0 -semigroups and C_0 -evolution families admit an (approximate) invertible dilation?”

They explain just above and below the question that the useful version for maximal inequalities requires the dilation spaces to retain the relevant geometry, such as Hilbertianity, 2-smoothness, or UMD.

s in Banach spaces are avail-
 MD Banach spaces [NVW07,
 . Each of the two approaches
 on of the stochastic integral in
 use in the theory of stochastic
 ability covers only half of the
 rz97]. It replaces the basic Itô
 entails some loss in precision.
 l regularity, which cannot be
 in UMD Banach spaces covers
 l leads to two-sided estimates
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 nly, but in that setting a full-

Definition from the Source

Let $(S(t, s))_{0 \leq s \leq t \leq T}$ be a C_0 -evolution family on a Banach space X . In the source terminology, it admits an invertible dilation on a Banach space Y if there are strongly continuous functions

$$J : [0, T] \rightarrow \mathcal{L}(X, Y), \quad Q : [0, T] \rightarrow \mathcal{L}(Y, X)$$

such that

$$S(t, s) = Q(t)J(s), \quad 0 \leq s \leq t \leq T.$$

Idea of the Proof

The factorization can be forced by remembering the future orbit of x from the starting time s . Put this orbit into the path space $C([0, T]; X)$, and make it constant before time s so that the path is continuous at the switching point. Evaluation at t then recovers $S(t, s)x$. Strong continuity of the maps $J(s)$ follows from the uniform strong continuity of the evolution family on the compact time triangle.

Theorem 1 (Formal path dilation). *Let X be a Banach space and let $(S(t, s))_{0 \leq s \leq t \leq T}$ be a C_0 -evolution family on X . Then $(S(t, s))$ admits an exact invertible dilation, in the literal sense of the source definition, on $Y = C([0, T]; X)$.*

Proof. Let $Y = C([0, T]; X)$ with the supremum norm. For $t \in [0, T]$, define

$$Q(t)f = f(t), \quad f \in Y.$$

Then $\|Q(t)\| \leq 1$, and $t \mapsto Q(t)$ is strongly continuous because each $f \in C([0, T]; X)$ is uniformly continuous.

For $s \in [0, T]$ and $x \in X$, define $J(s)x \in Y$ by

$$(J(s)x)(r) = \begin{cases} x, & 0 \leq r \leq s, \\ S(r, s)x, & s \leq r \leq T. \end{cases}$$

This is a continuous X -valued path, since $S(s, s) = I$. The family $\{S(t, s) : 0 \leq s \leq t \leq T\}$ is pointwise bounded by strong continuity on the compact triangle, hence uniformly bounded by the uniform boundedness principle. Therefore

$$\|J(s)x\|_\infty \leq \max \left\{ 1, \sup_{0 \leq a \leq b \leq T} \|S(b, a)\| \right\} \|x\|,$$

so $J(s) \in \mathcal{L}(X, Y)$ with a bound uniform in s .

It remains to check strong continuity of $s \mapsto J(s)$. Fix $x \in X$ and let $s_n \rightarrow s$. The map

$$(r, a) \mapsto S(r, a)x$$

is continuous on the compact triangle $0 \leq a \leq r \leq T$, hence uniformly continuous there. If $r \geq \max\{s_n, s\}$, uniform continuity gives

$$\|S(r, s_n)x - S(r, s)x\| \rightarrow 0$$

uniformly in r . If $r \leq \min\{s_n, s\}$, the two paths are both equal to x . On the remaining interval between s_n and s , one path is equal to x , while the other is $S(r, a)x$ with $|r - a| \leq |s_n - s|$; uniform continuity at the diagonal, together with $S(a, a) = I$, gives

$$\sup_{|r-a| \leq |s_n-s|} \|S(r, a)x - x\| \rightarrow 0.$$

Thus $\|J(s_n)x - J(s)x\|_\infty \rightarrow 0$, proving strong continuity.

Finally, for $0 \leq s \leq t \leq T$,

$$Q(t)J(s)x = (J(s)x)(t) = S(t, s)x.$$

This is the desired exact dilation. □

Remark 1 (Why this does not settle the geometric problem). *The construction generally destroys the geometry needed in the source's dilation theorem. If $X \neq \{0\}$, choose $x_0 \in X$ with $\|x_0\| = 1$. The map $f \mapsto (r \mapsto f(r)x_0)$ embeds $C([0, T])$ isometrically into $C([0, T]; X)$. Hence $C([0, T]; X)$ is non-reflexive. In particular it cannot be 2-smooth or UMD, since those classes are reflexive. It is also not a Hilbert space except in the trivial zero-dimensional case.*

Thus Theorem 1 answers the literal non-geometric version of the dilation question, but the source's useful question remains: when can such a factorization be arranged inside a class preserving Hilbertian, 2-smooth, UMD, or other stochastic-integration geometry?

Verification and Novelty Check

Local indexes were searched for arXiv:2006.08325 and for the phrases *invertible dilation*, *approximate invertible dilation*, *evolution family*, and *stochastic convolution*. No existing packet recorded this formal path-space factorization. Web searches on June 20, 2026 for combinations of *invertible dilation*, $C([0, T]; X)$, and *evolution family* did not find a later answer to the source question.

Human Review Recommendation

Review should treat this as a clarification packet, not as a full stochastic maximal-inequality solution. The proof of the factorization is elementary; the main review point is whether the run should record the source question as implicitly geometry-preserving. If so, this packet should remain in **partial/**: it identifies the formal obstruction and sharpens the remaining problem.

References

- [1] J. van Neerven and M. Veraar, *Maximal estimates for stochastic convolutions in 2-smooth Banach spaces and applications to stochastic evolution equations*, arXiv:2006.08325, 2020.